PREDICTING LLM OUTPUT LENGTH VIA ENTROPY-GUIDED REPRESENTATIONS

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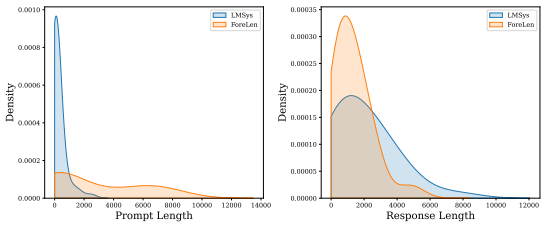
ABSTRACT

The long-tailed distribution of sequence lengths in LLM serving and reinforcement learning (RL) sampling causes significant computational waste due to excessive padding in batched inference. Existing methods rely on auxiliary models for static length prediction, but they incur high overhead, generalize poorly, and fail in stochastic "one-to-many" sampling scenarios. We introduce a lightweight framework that reuses the main model's internal hidden states for efficient length prediction. Our framework features two core components: 1) Entropy-Guided Token Pooling (EGTP), which uses on-the-fly activations and token entropy for highly accurate static prediction with negligible cost, and 2) Progressive Length Prediction (PLP), which dynamically estimates the remaining length at each decoding step to handle stochastic generation. To validate our approach, we build and release ForeLen, a comprehensive benchmark with long-sequence, Chain-of-Thought, and RL data. On ForeLen, EGTP achieves state-of-the-art accuracy, reducing MAE by 29.16% over the best baseline. Integrating our methods with a length-aware scheduler yields significant end-to-end throughput gains. Our work provides a new technical and evaluation baseline for efficient LLM inference.

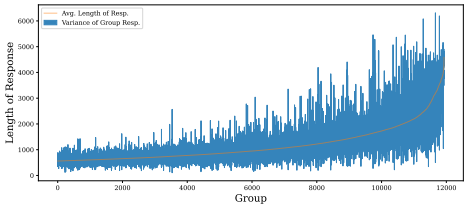
1 INTRODUCTION

In recent years, large language models (LLMs) (Achiam et al., 2023; Brown et al., 2020) have rapidly proliferated across diverse applications including chatbots (Yang et al., 2025a), code assistants (Petrovic et al., 2025; Cheng et al., 2025b), knowledge database (Zhang et al., 2024; Cheng et al., 2024), retrieval-augmented generation (RAG) (Li et al., 2025c; Yao et al., 2025; Cheng et al., 2025a; Ali et al., 2025) applications, and intelligent agents (Schmidgall et al., 2025). Supporting these applications is an efficient LLM serving infrastructure, and the typical LLM serving process follows an autoregressive paradigm: the system receives prompts generated by users or tasks, and the model constructs complete responses through iterative next-token prediction (Zhen et al., 2025; Kwon et al., 2023; Liu et al., 2025). Concurrently, LLM inference capabilities are being integrated into online reinforcement learning, such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024) and its variants (Yu et al., 2025; Zheng et al., 2025). To construct stable and diverse reward signals, systems must perform multiple independent sampling operations on the same prompt, generating a set of candidate responses and computing rewards based on their relative quality, which are then fed back to the policy update phases.

In real scenarios, batching techniques (Dong et al., 2025; Xuan et al., 2025) are the core mechanism for boosting hardware utilization and overall throughput. By executing multiple requests in parallel, systems can significantly amortize scheduling and memory-access overhead. However, the generation lengths of



(a) The prompt and response length distributions of the LMSYS dataset and our proposed ForeLen benchmark (left for prompt length, right for response length). The generation lengths of different requests within a batch typically exhibit highly variable distributions.



(b) Response length distribution observed during a GRPO training loop for online RL sampling. Despite using identical prompts within each group, stochastic decoding introduces substantial variability in generated response lengths, highlighting the challenge for dynamic length prediction.

Figure 1: **Analysis of LLM Sequence Length Distributions in Representative Scenarios.** Figure (a) illustrates the length characteristics of our proposed **ForeLen** benchmark and LMSYS, demonstrating a wider and longer-tailed distribution. Figure (b) showcases the high length variance for generations from identical prompts in an online RL training setting.

different requests within a batch usually vary greatly (as shown in Figure 1a). Because tensor shapes must align, shorter sequences are padded to match the longest one (Gururangan et al., 2020; Yun et al., 2024). This leads to a marked ”barrel effect”. Redundant padding computations enlarge GPU or accelerator time consumption (Deshmukh et al., 2025), and also reduce effective computation (Qiu et al., 2024; Piotrowski et al., 2025). If systems could estimate output lengths for each request before or early in inference (Han et al., 2025), they could apply length-aware scheduling. Such scheduling would cut ineffective computation. It would also raise throughput and cost efficiency in both online serving (Jin et al., 2023) and RL training (Zheng et al., 2025; Yu et al., 2025; Wang et al., 2025a; Li et al., 2025a;b; Zhang et al., 2025).

To predict output lengths for LLMs, recent studies (Qiu et al., 2024; Jin et al., 2023; Hu et al., 2024; Fu et al., 2024) typically attach a fine-tuned, lightweight auxiliary predictor, e.g., DistilBERT (Sanh et al., 2019) or OPT (Zhang et al., 2022). Although this design is appealing, it still suffers from three key limitations: (i) Instability in stochastic, “one-to-many” generation. During sampling—especially in reinforcement-learning workflows (Wang et al., 2025b)—a single prompt can yield multiple valid completions with widely differing lengths (Figure 1b). Any static estimate that relies only on the prompt therefore becomes unreliable. (ii) Limited accuracy and generalization. These predictors are usually trained on benchmarks such as LMSYS (Zheng et al., 2024), which contain few long sequences and little complex reasoning. As a result, their length forecasts deteriorate in realistic, more complex settings. (iii) Additional computational and deployment cost. Each request must run a separate predictor instead of reusing the rich hidden states already produced by the main LLM.

To overcome these challenges of inefficiency, generalization, and inflexibility, we propose a new framework that directly utilizes the information embedded within the LLM’s internal activations. Our core insight is that if an LLM can determine when to emit the <eos> token, then signals related to the eventual output length must be implicitly encoded in its internal states. By reusing these activations, we can enable more accurate length prediction with minimal additional cost. We introduce Entropy-Guided Token Pooling (EGTP), which reuses on-the-fly activations, guided by token entropy, to capture the most informative signals from the prompt, thereby addressing the overhead and generalization challenges of prior work.

Furthermore, to address the fundamental difficulty of length prediction in stochastic environments, we introduce Progressive Length Prediction (PLP). PLP leverages the autoregressive nature of LLMs by operating dynamically at each decoding step. It uses the current activations to produce an updated estimate of the

remaining tokens to be generated. By iteratively refining its forecast, PLP enables length-aware scheduling even in highly unpredictable environments like RL sampling, a task for which previous static methods are not inherently designed.

To rigorously validate our framework, particularly in scenarios where existing methods may struggle, we introduce ForeLen, the first comprehensive benchmark for length prediction featuring long-sequence, Chain-of-Thought (CoT), and reinforcement learning (RL) sampling data. Experiments on ForeLen show that our method, EGTP, achieves state-of-the-art accuracy. Averaged across all tested models, our method reduces the Mean Absolute Error (MAE) by 29.16% compared to the strongest baseline and 55.09% compared to the widely-used SSJF-Reg (Qiu et al., 2024). This superior prediction accuracy, in turn, yields significant improvements in end-to-end inference throughput.

Our contributions are summarized as follows:

- We propose a lightweight and efficient length prediction framework, comprising two core modules: **EGTP**, which reuses internal model activations guided by token entropy for accurate static prediction, and **PLP**, which performs progressive prediction to handle highly stochastic RL sampling.
- To facilitate rigorous evaluation, we construct and release **ForeLen**, the first comprehensive benchmark designed to test predictors on challenging long-sequence, CoT, and RL data.
- Our methods achieve state-of-the-art prediction accuracy on ForeLen and, when integrated with a length-aware scheduler, yield significant improvements in inference throughput.

2 Related Work

Efficient LLM Serving and Inference Optimization. Efficient LLM serving relies on optimizations like continuous batching (e.g., in vLLM) (Kwon et al., 2023) and efficient KV Cache management like PagedAttention (Kwon et al., 2023). These techniques maximize throughput by dynamically managing requests and mitigating memory fragmentation. However, while they reduce inter-request idle time, they do not solve the computational waste from padding within a running batch, known as the "barrel effect," where shorter sequences waste computation matching the longest (Deshmukh et al., 2025). Our work is orthogonal: by predicting output length, we enable length-aware schedulers to build more homogeneous batches, directly reducing this padding overhead and complementing existing serving architectures.

LLM Response Length Prediction. Prior work on length prediction primarily trains lightweight auxiliary models (e.g., DistilBERT) to predict an LLM's output length based only on the input prompt (Qiu et al., 2024; Jin et al., 2023; Hu et al., 2024). This approach, however, incurs non-negligible overhead, requiring a separate model to be trained, deployed, and executed for every request. We bypass this cost. Inspired by work linking internal model states like token entropy to generation structure (Li et al., 2025e), our EGTP (Entropy-Guided token Pooling) method reuses the LLM's own on-the-fly activations to achieve accurate prediction with negligible additional computation.

LLM Inference and Sampling in Reinforcement Learning. Modern LLM alignment algorithms like GRPO (Shao et al., 2024) and its variants (Yu et al., 2025; Zheng et al., 2025) require generating multiple candidate responses from the same prompt using stochastic sampling. This process creates extreme output length variance for a single input, rendering all existing static, prompt-based predictors (Qiu et al., 2024) completely ineffective. Furthermore, a standardized benchmark for this "one-to-many" prediction scenario is absent (Wang et al., 2025b). Our PLP (Progressive Length Prediction) is designed specifically for this dynamic setting. Instead of a single static forecast, it operates autoregressively, using current model activations at each step to iteratively update its prediction of the remaining tokens, thereby adapting to the unique path of each stochastic sample.

3 METHOD

Our proposed methodology tackles the challenge of length prediction through two primary components designed for static and dynamic scenarios, respectively. First, for static prediction from complex prompts, we introduce Entropy-Guided Token Pooling (EGTP). This method efficiently utilizes the LLM's internal activations, and its accuracy is further boosted by our novel Regression via Soft Label Distribution training strategy. Second, to handle the 'one-to-many' problem in stochastic environments like RL, we present Progressive Length Prediction (PLP), a dynamic approach that iteratively refines its forecast. This capability is crucial for scenarios unaddressed by prior static methods.

3.1 ENTROPY-GUIDED TOKEN POOLING (EGTP)

Motivation. As mentioned above, our core insight is that if an LLM can determine when to emit the <eos> token, then signals related to the eventual output length must be implicitly encoded in its internal states. By reusing these activations, we can enable accurate length prediction with minimal additional cost. However, a challenge arises because we have a sequence of these representations, but we need to produce a single predictive output. Therefore, we require a pooling mechanism to aggregate them into one conclusive vector. We find that traditional methods like mean or max pooling are suboptimal, as they can dilute or discard crucial information. Instead, we propose a novel pooling strategy guided by predictive entropy, which we believe more effectively captures the most informative tokens to create a superior final representation.

To validate this hypothesis, we measured each token's contribution to the final prediction using a **gradient-based attribution** method (Sundararajan et al., 2017). Specifically, we define the importance I_t of a token at position t as the L2 norm of the gradient of the Mean Squared Error (MSE) loss ($\mathcal{L}_{\text{MSE}}$) between the predicted and ground truth lengths, with respect to its hidden state representation h_t:

$$I_t = \|\nabla_{h_t} \mathcal{L}_{\text{MSE}}\|_2. \tag{1}$$

To provide an empirical foundation for our approach, we analyzed the relationship between token entropy and importance. The experiment was performed using a BERT model with a linear head on 10k samples randomly selected from the LMSYS dataset. As shown in Figure 2, the results demonstrate a significant positive correlation between a token's entropy and its importance for length prediction (Pearson's $r = 0.451$). This finding confirms that high-entropy tokens, which are those where the model is most uncertain about what to generate next, are indeed critical signals for forecasting the output length, thus validating our entropy-weighted pooling strategy.

EGTP. Based on this validation, we propose **Entropy-Guided Token Pooling (EGTP)**. This method aggregates a sequence of input hidden states $\{h_1, h_2, \ldots, h_n\}$ into a single feature vector $\mathbf{h}$. The process begins by computing the entropy H_i for each token. Specifically, for each hidden state h_i corresponding to an input token x_i, we calculate the entropy from the next-token probability distribution $P(v|x_{<i})$ over the vocabulary V:

$$H_i = -\sum_{v \in V} P(v|x_{<i}) \log P(v|x_{<i}) \tag{2}$$

Next, these entropy values are used to generate attention weights. We transform the entropies into a distribution of weights w_i via a softmax function, scaled by a temperature parameter α that controls the distribution's sharpness:

$$w_i = \frac{\exp(H_i)}{\sum_{j=1}^{n} \exp(H_j)} \tag{3}$$